Benchmarking Fisheries Integrated Modeling System: A Next-Generation Framework for Reliable Fisheries Stock Assessments

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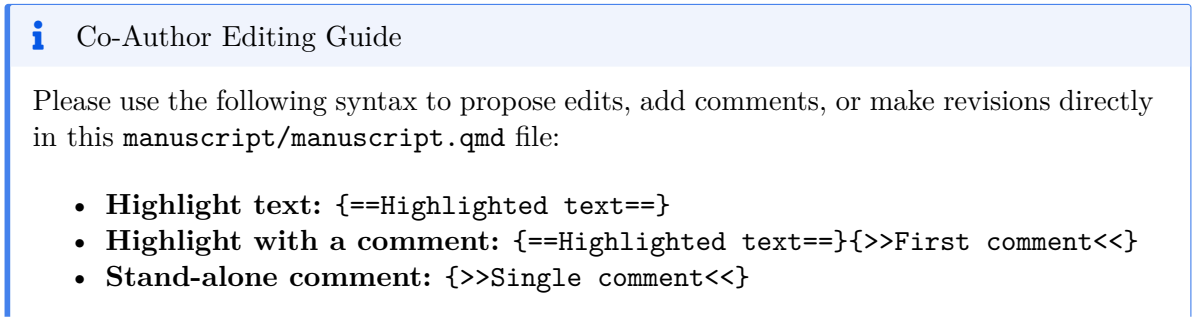
As NOAA Fisheries transitions to next-generation stock assessment methods, including the Fisheries Integrated Modeling System (FIMS), it is important to evaluate how its performance compares with established modeling tools. Using an age-structured operating model, we compared FIMS estimates with four widely used assessment models in the United States. Comparisons with ADMB models, the Age Structured Assessment Program, Stock Synthesis, and the Beaufort Assessment Model, focused on verifying estimates of key assessment quantities, including spawning biomass, recruitment, and fishing mortality. Comparisons with the Woods Hole Assessment Model focused on evaluating the performance of random-effects configurations. Results indicate that estimates from FIMS are highly consistent with those from established modeling frameworks, with median relative error closely aligned across key metrics. Differences among models were generally minor and primarily associated with differences in the sum-to-zero constraint on recruitment deviations, equilibrium initialization for initial numbers-at-age, and recruitment bias correction. Overall, these findings demonstrate that FIMS provides a reliable and modular framework for U.S. fisheries stock assessment while maintaining high standards of estimation accuracy and enabling flexible random-effects modeling.



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Introduction

Population-dynamics models are commonly used to assess the status of marine species and the sustainability of management decisions (TODO: Hilborn and Walter, 1992). Modernizing the models involves including adding the capacity to estimate random effects (TODO: find citation), fit to recapture data (e.g., close-kin mark-recapture data; TODO: citation), and use machine learning (e.g., random forests; TODO: citation maybe Caitlin's). Development of the latter two options are still in the exploratory stages but the inclusion of random effects is seen as a best practice, and has been for several years (TODO: citation).

With the development of the highly-used R package (R Core Team, 2026 TODO: add to bib file) Template Model Builder (TMB; TODO: Kristensen et al., 2016), the estimation of random effects is no longer limited to high-performance computers or simple linear models. In fact, several state-space population-dynamics models exist and are used for management (e.g., State-space Assessment Model, SAM; Woods Hole Assessment Model, WHAM; Climate-Enhanced, Age-based model with Temperature-specific Trophic Linkages and Energetics, RCEATTLE). - Laplace approximation - Benefits of random effects in population-dynamics models - Li et al. (2026) https://doi.org/10.1139/cjfas-2025-0093

In the United States, WHAM (Stock and Miller (2021)) was the first generalizable population-dynamics model written in TMB, where it is meant to replace the use of the Age-Structured Assessment Program (ASAP; TOOD: citation), a model used on the east coast. - NOAA's strategic plan / CAPAM - NOAA Fisheries is developing FIMS as a next-generation, extensible stock assessment framework.

Previous studies showed that the Age Structured Assessment Program (ASAP), Beaufort Assessment Model (BAM), and Stock Synthesis (SS3) produced similar results when configured similarly (Li et al. (2021)). All three of these frameworks are written in AD Model Builder (ADMB; TODO: citation), a statistical platform that has since discontinued in development, and thus, they cannot easily be modified to include next-generation features. In this manuscript we take the results of Li et al. (2021) one step further and investigate how legacy assumptions hold up when random effects are capable. Results verify estimates from the FIMS model by comparing them to both legacy, ADMB-based frameworks (i.e., ASAP, BAM, and SS3) and a modern TMB framework (i.e., WHAM).

Methods

- An age-structured operating model (OM) with grouper-like life history was used to simulate data with a 30-year duration (Li et al. (2021)).

Population abundance-at-age in year y and age a ($N_{a,y}$) is governed by exponential decay:

$$N_{a+1,y+1} = N_{a,y}e^{-Z_{a,y}} \quad \text{for } 1 \leq a < A-1$$

$$N_{A,y+1} = N_{A-1,y}e^{-Z_{A-1,y}} + N_{A,y}e^{-Z_{A,y}}$$

where $Z_{a,y} = M_a + F_{a,y}$ represents total instantaneous mortality, with age-invariant natural mortality $M = 0.20$ year^{-1} across all years and age classes (Li et al. 2021). Annual fishing mortality was modeled as separable into age-specific fishery selectivity ($S_{F,a}$) and fully selected annual fishing mortality (F_y): $F_{a,y} = S_{F,a}F_y$ (Li et al. 2021). The trajectory of F_y increased linearly over the 30-year time horizon from an initial rate of $F_1 = 0.01$ year^{-1} to a terminal rate of $F_{30} = 0.40$ year^{-1} (Li et al. 2021).

Recruitment (R_y, abundance at age 1) was governed by a standard Beverton–Holt spawner–recruit relationship parameterized by unfished equilibrium recruitment ($R_0 = 1.0 \times 10^6$ fish) and steepness ($h = 0.75$), with annual lognormal recruitment deviations:

$$R_y = \frac{0.8R_0hSSB_{y-1}}{0.2\phi_0R_0(1-h) + (h-0.2)SSB_{y-1}} e^{\varepsilon_{R,y} - \frac{\sigma_R^2}{2}}$$

where ϕ_0 is the unfished spawning biomass per recruit, SSB_y is the female spawning stock biomass at the time of spawning, and $\varepsilon_{R,y} \sim \mathcal{N}(0, \sigma_R^2)$ with recruitment variability $\sigma_R = 0.20$ (Li et al. 2021).

Data generated from the OM included annual fishery landings (in metric tons), a fishery-independent survey abundance index, and age-composition proportions for both the fishery and survey (Li et al. 2021). Observation error for fishery landings and survey indices was lognormally distributed with coefficients of variation $CV_L = 0.05$ and $CV_I = 0.20$, respectively (Li et al. 2021). Age composition data were simulated with multinomial observation error using an effective sample size of 200 samples per year for both fleets (Li et al. 2021). The full suite of operating model biological, fishery, and observation parameters is summarized in Table 1.

Table 1: Operating model (OM) life history, population dynamics, fishery, and observation error parameters (Li et al. 2021; Siegfried et al. 2016).

Parameter / Variable	Symbol	Value / Formulation	Estimated in EM
Model Dimensions			
Number of modeled years	Y	30 years ($y \in \{1, \ldots, 30\}$)	—
Number of age classes	A	Ages 1 to 12+ (plus group at 12)	—
Growth & Life History			
Asymptotic length	L_∞	800 mm	Fixed
Brody growth coefficient	K	0.18 year^{-1}	Fixed
Theoretical age at zero length	t_0	-1.36 years	Fixed
Length–weight scaling	θ_1	2.50×10^{-8} mt$\cdot$mm^{-3}	Fixed

Parameter / Variable	Symbol	Value / Formulation	Estimated in EM
Length–weight power	θ_2	3.0	Fixed
Natural mortality rate	M	0.20 year^{-1} (constant across ages and years)	Fixed
Sex ratio	r	0.50 (proportion female at all ages)	Fixed
Age at 50% maturity	$a_{50,\mathrm{mat}}$	2.25 years	Fixed
Slope of maturity ogive	β_{mat}	3.0	Fixed
Stock & Recruitment			
Unfished recruitment	R_0	1.0×10^6 fish (1.0×10^3 in thousands)	Estimated
Steepness	h	0.75	Fixed
Recruitment deviation standard deviation	σ_R	0.20 (log space)	Setting-dependent
Fishery & Survey			
Fully selected fishing mortality trajectory	F_y	Linear increase: $F_1 = 0.01$ to $F_{30} = 0.40$ year^{-1}	Estimated (F_y or apical F)
Fishery selectivity inflection (a_{50})	$a_{50,F}$	2.0 years (logistic)	Estimated
Fishery selectivity slope	β_F	1.0	Estimated
Survey selectivity inflection (a_{50})	$a_{50,I}$	1.5 years (logistic)	Estimated
Survey selectivity slope	β_I	2.0	Estimated
Survey catchability coefficient	q	3.46×10^{-7} (survey in numbers)	Estimated
Observation Uncertainty			
Fishery landings error	CV_L	0.05 (lognormal, mt)	Fixed
Survey index error	CV_I	0.20 (lognormal, numbers)	Fixed
Fleet age composition sample size	N_L	200 fish year^{-1} (multinomial)	Fixed

Parameter / Variable	Symbol	Value / Formulation	Estimated in EM
Survey age composition sample size	N_I	200 fish year^{-1} (multinomial)	Fixed

Estimation models and configurations

- ASAP, BAM, SS3, FIMS, Rceattle, and WHAM were configured as estimation models (EMs). They represent both established operational software and next-generation frameworks (Li et al. 2021):
  - **Age Structured Assessment Program (ASAP, version 3.0.16)**: An ADMB-based statistical catch-at-age model widely applied in the Northeast U.S. (Legault and Restrepo 1998; Li et al. 2021).
  - **Beaufort Assessment Model (BAM)**: An ADMB-based statistical catch-at-age framework developed by the Southeast Fisheries Science Center (Williams and Shertzer 2015; Li et al. 2021).
  - **Stock Synthesis (SS3, version 3.30.15)**: A flexible, generalized age- and length-structured assessment framework implemented in ADMB and used globally (Methot and Wetzel 2013; Li et al. 2021).
  - **Fisheries Integrated Modeling System (FIMS)**: NOAA's next-generation modular framework implemented in C++ and TMB [3, 6, 23].
  - **Rceattle**: A TMB age-structured framework developed for single-species and multi-species bioenergetic stock assessments in Alaska (Adams et al. 2022).
  - **Woods Hole Assessment Model (WHAM)**: A state-space age-structured stock assessment framework implemented in TMB that incorporates environmental linkages and random-effects structures (Stock and Miller 2021).
- All models were configured with equivalent population and fishery structures: ages 1–12+ with a plus group at age 12, a single fishing fleet with logistic selectivity, a single survey with logistic selectivity, annual lognormal observation likelihoods for landings ($CV = 0.05$) and survey index ($CV = 0.20$), and annual multinomial likelihoods for age composition data ($N = 200$) (Li et al. 2021, 8, 43, 46, 67, 92). Survey catchability q was estimated as a free parameter across all models [8, 33-37].

Table 2: Structural characteristics and benchmark parameter configurations across the six evaluated estimation models (EMs) [6, 8, 10-16, 20, 24, 38, 40].

Dimension / Setting	ASAP	BAM	SS3	FIMS	Rceattle	WHAM
Software Platform	ADMB	ADMB	ADMB	TMB	TMB	TMB
Initial Age Structure (N_1)	Equilibrium offset with initial F_1 [20]	Equilibrium offset with initial F_1 [20]	Equilibrium offset with initial F_1 [20]	Free parameters (`log_init_naa`) [11, 24]	Free parameters (`initMode = 0`) [11, 17]	Free parameters (`N1_model = 0`) [11, 17]
Stock-Recruit Formulation	Beverton–Holt (R_0, h) [8, 40]	Beverton–Holt (R_0, h) [8, 40]	Beverton–Holt (R_0, h) [8, 40]	Beverton–Holt (R_0, h) [12, 24]	Beverton–Holt $(\alpha, \beta$; α fixed from $h)$ [11, 12]	Beverton–Holt $(\alpha, \beta$; α fixed from $h)$ [11, 14]
Fishing Mortality (F)	Apical F with annual devs	Apical F with annual devs	Continuous apical hybrid F	Annual $\log F_{\text{mort}}$ [27, 36]	Annual $\log F$ [42, 43]	Annual $\log F_{\text{tot}}$ [14, 44]
Survey Catchability (q)	Free parameter ($\log q$) [8, 35]	Free parameter ($\log q$) [8, 44]	Free parameter ($\log q$) [8, 44]	Free parameter ($\log q$) [27, 36]	Free parameter (closed-form MLE) [33, 37, 43]	Free parameter ($\log q$) [14, 44]
Recruitment: PML	Quadratic penalty ($\sigma_R = 0.20$) [10]	Quadratic penalty ($\sigma_R = 0.20$) [45]	Quadratic penalty ($\sigma_R = 0.20$) [13]	Penalized fixed effects (`log_sd` fixed) [31]	Penalized fixed effects (`random_rec = FALSE`) [11, 12]	Penalized fixed effects (`log_NAA` removed from RE) [16]
Recruitment: State-Space (est. σ_R)	Not supported	Not supported	Not supported	Random effects (`log_sd` estimated) [24]	Random effects (`R_log_sd` estimated) [11, 12]	Random effects (`log_NAA_sigma` estimated) [13, 14]

Dimension / Setting	ASAP	BAM	SS3	FIMS	Rceattle	WHAM
Recruitment: State-Space (known σ_R)	Not supported	Not supported	Not supported	Random effects (`log_sd` mapped constant) [30]	Random effects (`R_log_sd` mapped to `NA`) [11, 12]	Random effects (`log_NAA_sigma` mapped to `NA`) [15]

Recruitment Modeling and Random-Effects Scenarios

Recruitment deviations in statistical catch-at-age models can be treated as penalised fixed effects or as continuous random variables (Li et al. 2021; Stock and Miller 2021, 45, 165). To rigorously test how recruitment configuration influences estimation performance, the three modern TMB frameworks (FIMS, Rceattle, and WHAM) were each evaluated across three distinct recruitment scenarios [11, 17, 46, 47]:

- **Penalized Maximum Likelihood (PML) Scenario (`penalized_MLE` / `fixed_effects`)**: Recruitment deviations ($\varepsilon_{R,y}$) are estimated as fixed-effect parameters subject to a Gaussian log-likelihood penalty:

$$\mathcal{L}_{\text{pen}}(\varepsilon_R) = \frac{1}{2\sigma_R^2} \sum_{y=2}^{Y} \varepsilon_{R,y}^2$$

  The standard deviation of recruitment deviations (σ_R) was fixed at the true operating model value ($\sigma_R = 0.20$) [10, 11, 13, 31, 45]. This matches the formulation used in classical ADMB models (ASAP, BAM, SS3) [10, 11, 13, 45]. In FIMS, this was configured by setting the recruitment `log_devs` parameter to `estimation_type = "fixed_effects"` with `log_sd` held constant [31]. In Rceattle, `random_rec = FALSE` was set in `Rceattle::fit_mod`, which maps off `R_log_sd` [11, 12]. In WHAM, `log_NAA` was stripped from the internal random-effects vector (`input_wham[["random"]]`) [16].

- **State-Space with Unknown Recruitment Standard Deviation (`state_space_unknown_sigmaR`)**: Recruitment deviations are treated as unobserved latent state variables (random effects) integrated out of the joint likelihood via the Laplace approximation in TMB [11, 14, 24-26]. The variance parameter σ_R (or $\log \sigma_R$) is simultaneously estimated as a fixed-effect hyperparameter [11, 13, 14, 24]. In FIMS, this corresponds to default random effects estimation where `log_sd` is estimated freely [24, 28]. In Rceattle, `random_rec = TRUE` was activated with `R_log_sd` estimated freely [11, 12]. In WHAM, `NAA_re = list(sigma = "rec")` was specified with `log_NAA_sigma` estimated [13, 14].

- **State-Space with Known Recruitment Standard Deviation (`state_space_known_sigmaR`)**: Recruitment deviations remain random effects integrated via the Laplace approximation, but the variance hyperparameter σ_R is fixed at the operating model's true value ($\sigma_R = 0.20$) [11, 12, 15, 30]. In FIMS, `log_sd` was set to `estimation_type = "constant"` [30]. In Rceattle, the parameter mapping vector was modified such that `mapFactor$R_log_sd <- factor(rep(NA, ...))` [11, 12]. In WHAM, `input_wham[["map"]][["log_NAA_sigma"]]` was mapped to `factor(NA)` [15].

- A total of 500 simulation replicates were initially generated to ensure sufficient sample size for convergence analysis. The final analysis was performed on a subset of 100 simulations that successfully converged across all models.

  - Convergence was evaluated using two criteria:

    * The optimization algorithm terminated with a maximum absolute gradient of the log-likelihood (or marginal log-likelihood) less than or equal to a strict threshold: $\max|\nabla\ell| \leq 0.004$.
    * The Hessian matrix at the final solution was verified to be strictly positive-definite, confirming that the optimizer located a local minimum and standard errors were calculable.

  - Across the 500 simulations, the convergence rate (CR_m) for model m was calculated as:

    $$CR_m = \frac{N_{\text{converged},m}}{N_{\text{total},m}} \times 100$$

    To guarantee direct, paired comparisons across all model configurations, we identified the common intersection of simulation replicates that converged across all estimation models. In addition, simulations exhibiting catastrophic estimation failures across any EM (defined as absolute relative error $|RE| > 40\%$) were filtered out as numerical outliers [55]. The first 100 fully converged, non-outlier simulations common to all models were selected for the final comparative evaluation.

- Model performance metrics included:

  - Median relative error (MRE) and median absolute relative error (MARE) for key quantities such as spawning biomass (SB), recruitment (R), and fishing mortality (F).

    * Relative error (RE) was calculated as $\frac{\hat{\theta}_i}{\theta_i} - 1$, where θ_i is the true value for simulated dataset i and $\hat{\theta}_i$ is the value estimated from fitting the model to the simulated data.
    * 95% confidence intervals of the MRE were calculated using the binomial distribution to estimate bias of each EM for a given OM (Thompson (1936)).

    - To summarize the bias across simulations and years for each model, we calculated the median RE across years for each simulation and then took the mean of these medians across simulations (Stock and Miller (2021)).
    - MARE across all years and simulations was calculated to quantify the overall accuracy, providing a measure of the magnitude of estimation errors.
  - The uncertainty in estimates was assessed using the standard deviation of the REs across simulations for each year and EM, providing insight into the variability of model performance over time.
  - The run time for each estimation model was recorded to evaluate computational efficiency and compare the relative speed of different EMs across simulation replicates. Separating parameter optimization from Hessian calculation and uncertainty reporting (`sdreport`) allows for a more detailed assessment of computational efficiency.

Results

- Across the 500 initial operating model (OM) simulations, model convergence rates (CR) reflected differences in software framework implementations and the treatment of recruitment deviations (Table 3). TMB models generally exhibited higher values than ADMB models under the specified convergence criteria. FIMS, Rceattle, and WHAM showed particularly strong convergence when configured with random effects. However, these results are based on a one-size-fits-all criterion and may change if model-specific gradient thresholds are applied.

  - Stock Synthesis (SS3) and all three configurations of FIMS (`penalized_MLE`, `state_space_unknown_sigmaR`, and `state_space_known_sigmaR`) achieved a 100% convergence rate (500/500 replicates), satisfying both the gradient threshold ($\max|\nabla\ell| \leq 0.004$) and the positive-definite Hessian criterion. BAM converged in 498 of 500 replicates (99.6%), and ASAP converged in 480 replicates (96.0%).

  - For the other TMB-based platforms, convergence characteristics varied by recruitment formulation:

    - In WHAM, both state-space random-effects configurations converged in 100% of replicates (500/500), while the penalized fixed-effects configuration (`WHAM_penalized_MLE`) converged in 73.0% of replicates.
    - In Rceattle, state-space configurations achieved convergence rates of 96.2% (unknown σ_R) and 95.2% (known σ_R), while the penalized maximum likelihood configuration converged in 87.8% of replicates.

Table 3: Convergence success for 500 simulations across seven estimation model (EM) configurations. Convergence was defined as a positive-definite Hessian matrix and a maximum gradient component threshod.

X	model	total_simulations	converged_simulations	convergence_rate
1	ASAP	500	480	96.0
2	BAM	500	498	99.6
3	SS3	500	500	100.0
4	FIMS_penalized_MLE	500	500	100.0
5	FIMS_state_space_unknown_sigmaR	500	500	100.0
6	FIMS_state_space_known_sigmaR	500	500	100.0
7	Rceattle_penalized_MLE	500	439	87.8
8	Rceattle_state_space_unknown_sigmaR	500	481	96.2
9	Rceattle_state_space_known_sigmaR	500	476	95.2
10	WHAM_penalized_MLE	500	365	73.0
11	WHAM_state_space_unknown_sigmaR	500	500	100.0
12	WHAM_state_space_known_sigmaR	500	500	100.0

- Time-series estimates from all EMs captured the overall trends observed in the operating model, and the variability across the 100 simulations was consistent among EMs and closely matched the variability of the OM (Figure 1).

- Across all EMs, the median RE remained close to zero, indicating minimal bias in estimates relative to the OM. The distributions of REs showed consistent patterns among EMs over time (Figure 2). In the early years, REs were generally small and narrowly distributed, reflecting high precision in model estimates. In later years, the distributions widen across all EMs, indicating increased variability in the estimates. Overall, both the central tendency and variability of REs were similar across EMs, with FIMS configurations performing comparably to the other EMs (Figure 3).

- The mean of median RE for FIMS showed strong consistency with other EMs. Across all metrics, FIMS values were within the same decimal range as the others. In addi-

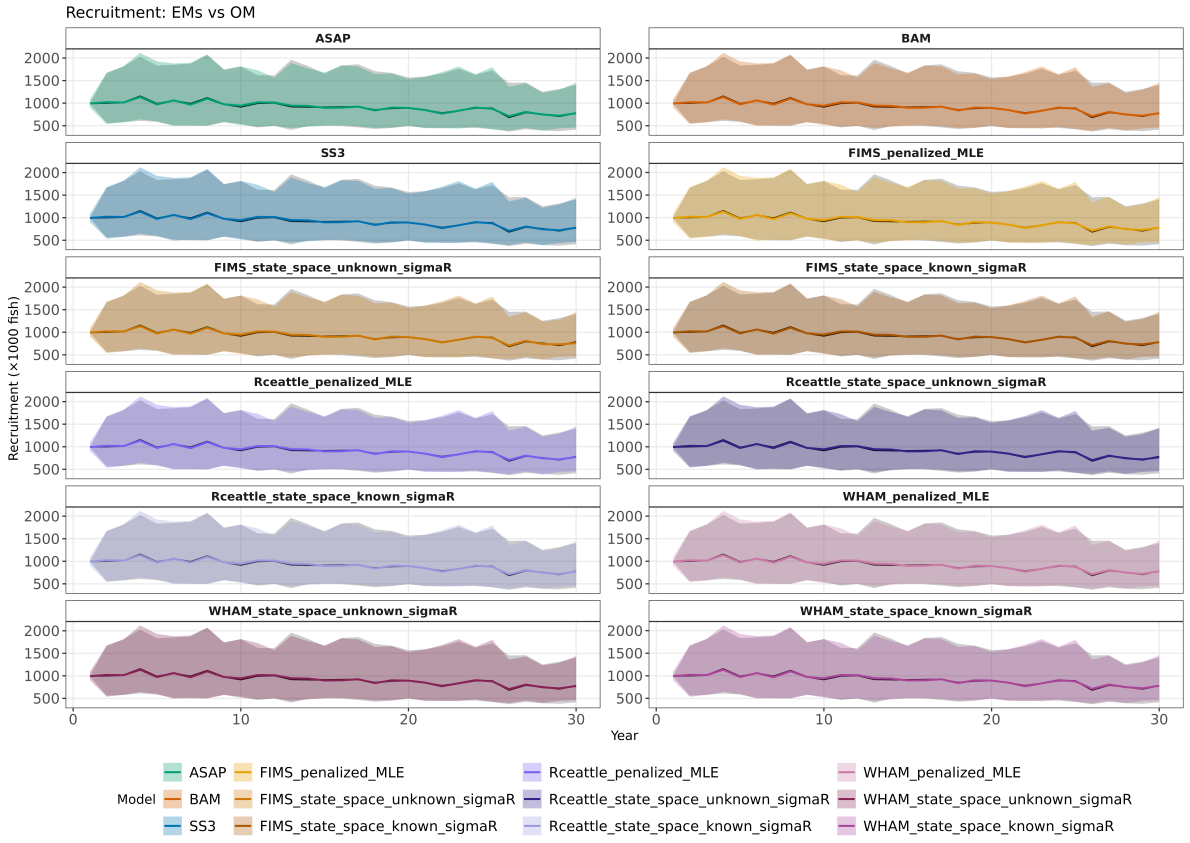

Figure 1: Time series of recruitment estimates from each estimation model (EM; ASAP, BAM, SS3, WHAM with fixed effects, WHAM with random effects, FIMS with fixed effects, and FIMS with random effects) compared to the true values from the operating model (OM). Colored lines represent the median recruitment across 100 simulation replicates for each EM, and colored shaded areas indicate the 95% confidence intervals. The black line in the center shows the true median recruitment from the OM, with the gray shaded area representing the OM's 95% confidence interval.

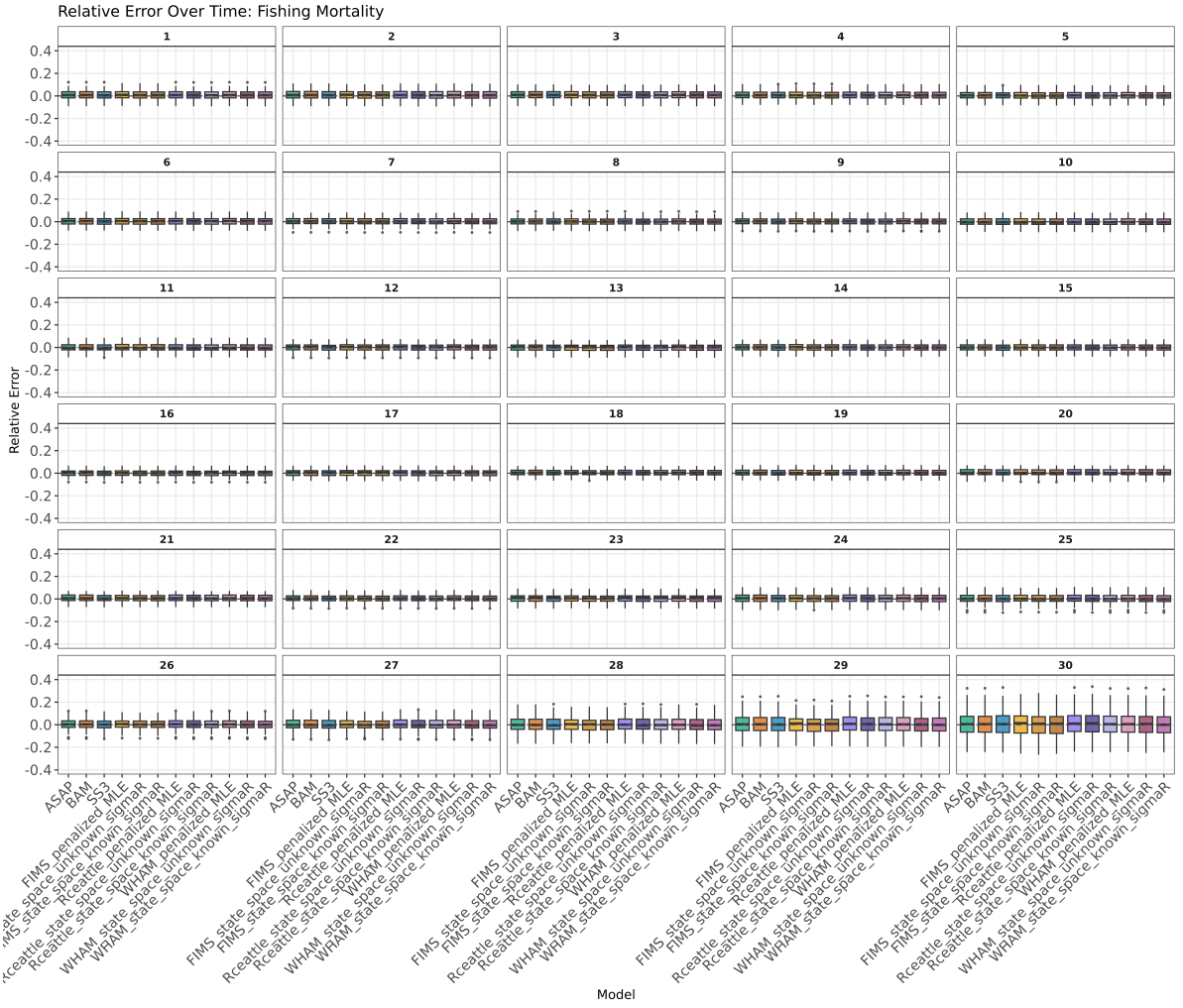

Figure 2: Relative error (RE) in fishing mortality over time from each estimation model (EM). Each facet represents one year from 1 to 30, and within each facet, seven boxplots show REs across 100 simulation replicates for the EMs: ASAP, BAM, SS3, WHAM with fixed effects, WHAM with random effects, FIMS with fixed effects, and FIMS with random effects.

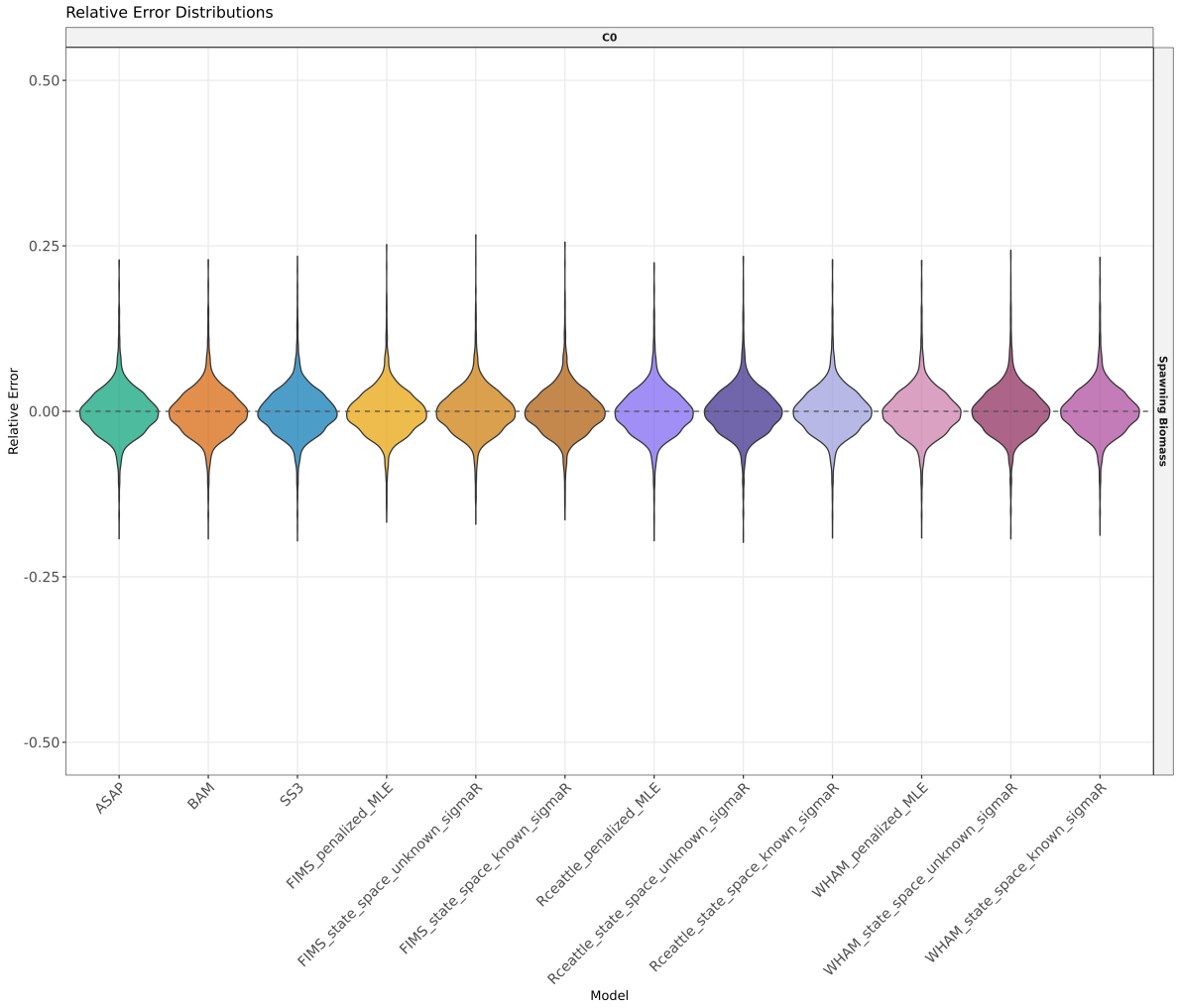

Figure 3: Distribution of relative error (RE) in spawning biomass across all years and 100 simulation replicates for each estimation model (EM). Violin plots summarize the full distribution of REs for ASAP, BAM, SS3, WHAM with fixed effects, WHAM with random effects, FIMS with fixed effects, and FIMS with random effects.

tion, random-effects models consistently reduced bias compared to their fixed-effects counterparts across all models.

Table 4: Mean of Median Relative Error (MRE; %) for key assessment metrics. 95% confidence intervals (mre_lower, mre_upper) and standard deviation (sd_mre) are calculated across 100 converged simulation replicates.

case	model	metric	mre	mre_lower	mre_upper	sd_mre	n_sims
C0	ASAP	fishing_mortality	0.41	-0.40	0.96	2.98	100
C0	BAM	fishing_mortality	0.41	-0.41	0.97	2.99	100
C0	SS3	fishing_mortality	0.26	-0.55	0.87	2.95	100
C0	FIMS_penalized_MLE	fishing_mortality	0.44	-0.23	0.97	3.06	100
C0	FIMS_state_space_unknown_sigmaR	fishing_mortality	0.29	-0.47	0.87	3.07	100
C0	FIMS_state_space_known_sigmaR	fishing_mortality	0.28	-0.42	0.82	3.05	100
C0	Rceattle_penalized_MLE	fishing_mortality	0.42	-0.41	0.98	2.99	100
C0	Rceattle_state_space_unknown_sigmaR	fishing_mortality	0.27	-0.55	0.90	3.01	100
C0	Rceattle_state_space_known_sigmaR	fishing_mortality	0.26	-0.52	0.82	2.99	100
C0	WHAM_penalized_MLE	fishing_mortality	0.41	-0.41	0.96	2.99	100
C0	WHAM_state_space_unknown_sigmaR	fishing_mortality	0.26	-0.58	0.86	3.00	100
C0	WHAM_state_space_known_sigmaR	fishing_mortality	0.25	-0.59	0.84	2.99	100
C0	ASAP	recruitment	0.19	-0.23	0.46	1.53	100
C0	BAM	recruitment	0.19	-0.25	0.46	1.53	100
C0	SS3	recruitment	0.23	-0.16	0.55	1.53	100
C0	FIMS_penalized_MLE	recruitment	0.20	-0.27	0.47	1.53	100
C0	FIMS_state_space_unknown_sigmaR	recruitment	0.26	-0.02	0.52	1.51	100
C0	FIMS_state_space_known_sigmaR	recruitment	0.24	-0.24	0.51	1.53	100

Table 4: Mean of Median Relative Error (MRE; %) for key assessment metrics. 95% confidence intervals (mre_lower, mre_upper) and standard deviation (sd_mre) are calculated across 100 converged simulation replicates.

case	model	metric	mre	mre_lower	mre_upper	sd_mre	n_sims
C0	Rceattle_penalized_MLE	recruitment	0.13	-0.30	0.36	1.53	100
C0	Rceattle_state_space_unknown_sigmaR	recruitment	0.21	-0.26	0.51	1.51	100
C0	Rceattle_state_space_known_sigmaR	recruitment	0.18	-0.24	0.45	1.53	100
C0	WHAM_penalized_MLE	recruitment	0.19	-0.20	0.45	1.53	100
C0	WHAM_state_space_unknown_sigmaR	recruitment	0.25	-0.05	0.47	1.52	100
C0	WHAM_state_space_known_sigmaR	recruitment	0.24	-0.15	0.48	1.53	100
C0	ASAP	spawning_biomass	-0.11	-0.60	0.51	2.12	100
C0	BAM	spawning_biomass	-0.12	-0.60	0.53	2.12	100
C0	SS3	spawning_biomass	-0.03	-0.59	0.55	2.12	100
C0	FIMS_penalized_MLE	spawning_biomass	-0.15	-0.79	0.54	2.20	100
C0	FIMS_state_space_unknown_sigmaR	spawning_biomass	-0.01	-0.63	0.66	2.22	100
C0	FIMS_state_space_known_sigmaR	spawning_biomass	0.01	-0.61	0.70	2.21	100
C0	Rceattle_penalized_MLE	spawning_biomass	-0.14	-0.62	0.54	2.13	100
C0	Rceattle_state_space_unknown_sigmaR	spawning_biomass	0.01	-0.50	0.68	2.15	100
C0	Rceattle_state_space_known_sigmaR	spawning_biomass	0.03	-0.47	0.70	2.13	100
C0	WHAM_penalized_MLE	spawning_biomass	-0.12	-0.60	0.52	2.12	100
C0	WHAM_state_space_unknown_sigmaR	spawning_biomass	0.02	-0.49	0.65	2.14	100
C0	WHAM_state_space_known_sigmaR	spawning_biomass	0.04	-0.43	0.68	2.13	100

- Summary statistics of absolute relative error across all years and simulations showed similar performance among EMs for all quantities of interest (Table 5). For fishing mortality, FIMS performed comparably to the other models, with a lowest MARE of 2.63%. Recruitment estimates exhibited the largest errors across all models. However, the FIMS random-effects model produced the lowest MARE (3.82%), slightly outperforming ASAP, BAM, SS3, and WHAM (3.83%–3.87%). For spawning biomass, FIMS yielded lowest MARE values (2.08% and 2.11%), indicating high estimation accuracy relative to the OM. Overall, the error metrics for FIMS were similar to those of the established models across all quantities, demonstrating that FIMS achieved comparable estimation performance.

Table 5: Median absolute relative error (MARE; %) in fishing mortality, recruitment, and spawning biomass. Metrics include MARE with 95% confidence intervals (mare_lower and mare_upper), mean absolute relative error (mean_are), and the standard deviation of absolute relative error (sd_are) calculated across all years and 100 simulation replicates (n = 3000 observations per metric).

case	model	metric	mare	mare_lower	mare_upper	mean_are	sd_are	n_obs
C0	ASAP	fishing_mortality	2.56	NA	NA	3.27	3.08	3000
C0	BAM	fishing_mortality	2.57	2.43	2.67	3.28	3.08	3000
C0	SS3	fishing_mortality	2.57	2.45	2.69	3.25	3.07	3000
C0	FIMS_penalized_MLE	fishing_mortality	2.53	2.41	2.63	3.23	2.98	3000
C0	FIMS_state_space_unknown_sigmaR	fishing_mortality	2.51	2.40	2.64	3.22	3.00	3000
C0	FIMS_state_space_known_sigmaR	fishing_mortality	2.52	2.41	2.64	3.21	2.96	3000
C0	Rceattle_penalized_MLE	fishing_mortality	2.54	2.44	2.66	3.28	3.10	3000
C0	Rceattle_state_space_unknown_sigmaR	fishing_mortality	2.55	2.44	2.68	3.27	3.11	3000

Table 5: Median absolute relative error (MARE; %) in fishing mortality, recruitment, and spawning biomass. Metrics include MARE with 95% confidence intervals (mare_lower and mare_upper), mean absolute relative error (mean_are), and the standard deviation of absolute relative error (sd_are) calculated across all years and 100 simulation replicates (n = 3000 observations per metric).

case	model	metric	mare	mare_lower	mare_upper	mean_are	sd_are	n_obs
C0	Rceattle_state_space_known_sigmaR	fishing_mortality	2.55	2.44	2.67	3.26	3.07	3000
C0	WHAM_penalized_MLE	fishing_mortality	2.56	2.43	2.67	3.28	3.08	3000
C0	WHAM_state_space_unknown_sigmaR	fishing_mortality	2.57	2.44	2.67	3.27	3.08	3000
C0	WHAM_state_space_known_sigmaR	fishing_mortality	2.55	2.44	2.68	3.26	3.05	3000
C0	ASAP	recruitment	3.82	NA	NA	5.00	4.48	3000
C0	BAM	recruitment	3.82	3.65	3.96	5.00	4.48	3000
C0	SS3	recruitment	3.80	3.66	3.97	5.00	4.49	3000
C0	FIMS_penalized_MLE	recruitment	3.79	3.64	3.95	5.05	4.62	3000
C0	FIMS_state_space_unknown_sigmaR	recruitment	3.78	3.63	3.95	5.08	4.70	3000
C0	FIMS_state_space_known_sigmaR	recruitment	3.79	3.65	3.95	5.05	4.63	3000
C0	Rceattle_penalized_MLE	recruitment	3.79	3.65	3.97	4.99	4.45	3000
C0	Rceattle_state_space_unknown_sigmaR	recruitment	3.81	3.64	3.97	5.02	4.53	3000
C0	Rceattle_state_space_known_sigmaR	recruitment	3.80	3.65	3.96	4.99	4.46	3000
C0	WHAM_penalized_MLE	recruitment	3.81	3.65	3.96	5.00	4.48	3000
C0	WHAM_state_space_unknown_sigmaR	recruitment	3.81	3.66	3.97	5.03	4.56	3000
C0	WHAM_state_space_known_sigmaR	recruitment	3.81	3.64	3.97	5.00	4.50	3000

Table 5: Median absolute relative error (MARE; %) in fishing mortality, recruitment, and spawning biomass. Metrics include MARE with 95% confidence intervals (mare_lower and mare_upper), mean absolute relative error (mean_are), and the standard deviation of absolute relative error (sd_are) calculated across all years and 100 simulation replicates (n = 3000 observations per metric).

case	model	metric	mare	mare_lower	mare_upper	mean_are	sd_are	n_obs
C0	ASAP	spawning_biomass	2.05	1.94	2.14	2.60	2.47	3000
C0	BAM	spawning_biomass	2.05	1.95	2.13	2.61	2.47	3000
C0	SS3	spawning_biomass	2.02	1.95	2.12	2.60	2.48	3000
C0	FIMS_penalized_MLE	spawning_biomass	1.98	1.89	2.10	2.55	2.39	3000
C0	FIMS_state_space_unknown_sigmaR	spawning_biomass	1.99	1.89	2.10	2.56	2.42	3000
C0	FIMS_state_space_known_sigmaR	spawning_biomass	1.98	1.88	2.06	2.55	2.40	3000
C0	Rceattle_penalized_MLE	spawning_biomass	2.05	1.96	2.14	2.60	2.45	3000
C0	Rceattle_state_space_unknown_sigmaR	spawning_biomass	2.06	1.98	2.15	2.61	2.48	3000
C0	Rceattle_state_space_known_sigmaR	spawning_biomass	2.04	1.95	2.13	2.60	2.46	3000
C0	WHAM_penalized_MLE	spawning_biomass	2.05	1.95	2.14	2.61	2.47	3000
C0	WHAM_state_space_unknown_sigmaR	spawning_biomass	2.04	1.96	2.14	2.61	2.49	3000
C0	WHAM_state_space_known_sigmaR	spawning_biomass	2.03	1.95	2.12	2.60	2.48	3000

- To verify that model frameworks produced mathematically equivalent parameter variance-covariance structures, standard errors for derived quantities were compared across models (Table 6).
  - For spawning biomass, the mean estimated standard error was ~171 mt across all reporting models: FIMS estimated standard errors of 170.4 mt (PML) and 170.7–170.8 mt (state-space), which matched SS3 (171.6 mt), ASAP (171.8 mt), Rceattle (171.8–172.4 mt), and WHAM (172.0–172.4 mt). Estimated standard

errors for annual recruitment were likewise identical across all frameworks, averaging 54.0–55.0 thousand fish (mean: 60.0–61.3 thousand fish; standard deviation: 26.9–28.4 thousand fish). BAM does not report analytical standard errors for derived time series in its standard output routines and Rceattle does not report standard errors for fishing mortality.

Table 6: Estimated standard errors for spawning biomass ((SB), mt) and recruitment ((R), thousands of fish) across 100 converged simulation replicates ((n = 3,000) estimates per metric).

case	model	metric	total_estimates	recorded_sims	n_simulations	mean_se	median_se	sd_se	min_se	max_se
C0	ASAP	spawning_biomass_se	3000	3000	100	171.76	153.66	76.78	61.09	375.05
C0	BAM	spawning_biomass_se	3000	0	0	NA	NA	NA	NA	NA
C0	SS3	spawning_biomass_se	3000	3000	100	171.61	153.89	75.97	61.84	374.59
C0	FIMS_penalized_MLE	spawning_biomass_se	3000	3000	100	170.42	152.88	76.82	59.30	371.80
C0	FIMS_state_space_unknown_sigmaR	spawning_biomass_se	3000	3000	100	170.69	153.54	76.99	59.47	374.47
C0	FIMS_state_space_known_sigmaR	spawning_biomass_se	3000	3000	100	170.83	153.30	76.95	59.50	372.61
C0	Rceattle_penalized_MLE	spawning_biomass_se	3000	3000	100	171.79	153.70	76.85	61.03	375.49
C0	Rceattle_state_space_unknown_sigmaR	spawning_biomass_se	3000	2970	99	172.14	154.25	76.99	61.25	376.29
C0	Rceattle_state_space_known_sigmaR	spawning_biomass_se	3000	2970	99	172.43	154.24	76.97	61.25	376.29
C0	WHAM_penalized_MLE	spawning_biomass_se	3000	3000	100	171.97	153.86	76.92	61.07	375.54
C0	WHAM_state_space_unknown_sigmaR	spawning_biomass_se	3000	3000	100	172.28	154.11	77.09	61.29	376.31
C0	WHAM_state_space_known_sigmaR	spawning_biomass_se	3000	3000	100	172.40	154.28	77.04	61.29	376.32
C0	ASAP	recruitment_se	3000	3000	100	60.87	54.86	27.29	21.64	320.01

Table 6: Estimated standard errors for spawning biomass ((SB), mt) and recruitment ((R), thousands of fish) across 100 converged simulation replicates ((n = 3,000) estimates per metric).

case	model	metric	total_estimates	recorded_sims	n_simulations	mean_se	median_se	sd_se	min_se	max_se
C0	BAM	recruitment_se	3000	0	0	NA	NA	NA	NA	NA
C0	SS3	recruitment_se	3000	3000	100	59.97	54.04	27.70	21.63	317.52
C0	FIMS_penalized_MLE	recruitment_se	3000	3000	100	61.25	54.94	28.34	21.63	340.12
C0	FIMS_state_space_recruitknown_sigmaR	recruitment_se	3000	3000	100	61.18	54.87	28.19	21.61	353.91
C0	FIMS_state_space_recruitknown_sigmaR	recruitment_se	3000	3000	100	61.29	55.00	28.36	21.65	340.19
C0	Rceattle_penalized_MLE	recruitment_se	3000	3000	100	60.82	54.85	27.19	21.59	318.99
C0	Rceattle_state_space_unknown_sigmaR	recruitment_se	3000	2970	99	60.73	54.78	26.93	21.60	329.02
C0	Rceattle_state_space_known_sigmaR	recruitment_se	3000	2970	99	61.00	54.95	27.30	21.62	319.22
C0	WHAM_penalized_MLE	recruitment_se	3000	3000	100	60.91	54.91	27.34	21.63	321.27
C0	WHAM_state_space_recruitknown_sigmaR	recruitment_se	3000	3000	100	60.86	54.89	27.22	21.60	332.59
C0	WHAM_state_space_recruitknown_sigmaR	recruitment_se	3000	3000	100	60.96	54.95	27.37	21.66	321.51
C0	ASAP	abundance_se	3000	0	0	NA	NA	NA	NA	NA
C0	BAM	abundance_se	3000	0	0	NA	NA	NA	NA	NA
C0	SS3	abundance_se	3000	0	0	NA	NA	NA	NA	NA
C0	FIMS_penalized_MLE	abundance_se	3000	0	0	NA	NA	NA	NA	NA
C0	FIMS_state_space_abundknown_sigmaR	abundance_se	3000	0	0	NA	NA	NA	NA	NA

Table 6: Estimated standard errors for spawning biomass ((SB), mt) and recruitment ((R), thousands of fish) across 100 converged simulation replicates ((n = 3,000) estimates per metric).

case	model	metric	total_estimates	recorded_sims	n_simulations	mean_se	median_se	sd_se	min_se	max_se
C0	FIMS_state_space_known_sigmaR	abundance_se	3000	0	0	NA	NA	NA	NA	NA
C0	Rceattle_penalized_MLE	abundance_se	3000	0	0	NA	NA	NA	NA	NA
C0	Rceattle_state_space_unknown_sigmaR	abundance_se	3000	0	0	NA	NA	NA	NA	NA
C0	Rceattle_state_space_known_sigmaR	abundance_se	3000	0	0	NA	NA	NA	NA	NA
C0	WHAM_penalized_MLE	abundance_se	3000	3000	100	115.81	97.63	63.36	44.42	522.32
C0	WHAM_state_space_unknown_sigmaR	abundance_se	3000	3000	100	115.81	97.72	63.15	44.65	528.48
C0	WHAM_state_space_known_sigmaR	abundance_se	3000	3000	100	116.04	97.87	63.47	44.51	522.89
C0	ASAP	fishing_mortality_se	3000	3000	100	0.01	0.01	0.02	0.00	0.12
C0	BAM	fishing_mortality_se	3000	0	0	NA	NA	NA	NA	NA
C0	SS3	fishing_mortality_se	3000	3000	100	0.01	0.01	0.02	0.00	0.12
C0	FIMS_penalized_MLE	fishing_mortality_se	3000	3000	100	0.01	0.01	0.02	0.00	0.12
C0	FIMS_state_space_known_sigmaR	fishing_mortality_se	3000	3000	100	0.01	0.01	0.02	0.00	0.12
C0	FIMS_state_space_unknown_sigmaR	fishing_mortality_se	3000	3000	100	0.01	0.01	0.02	0.00	0.12

Table 6: Estimated standard errors for spawning biomass ((SB), mt) and recruitment ((R), thousands of fish) across 100 converged simulation replicates ((n = 3,000) estimates per metric).

case	model	metric	total_estimates	recorded_sims	n_simulations	mean_se	median_se	sd_se	min_se	max_se
C0	Rceattle_penalized_MLE	fishing_mortality_se	3000	0	0	NA	NA	NA	NA	NA
C0	Rceattle_state_space_unknown_sigmaR	fishing_mortality_se	3000	0	0	NA	NA	NA	NA	NA
C0	Rceattle_state_space_known_sigmaR	fishing_mortality_se	3000	0	0	NA	NA	NA	NA	NA
C0	WHAM_penalized_MLE	fishing_mortality_se	3000	3000	100	0.01	0.01	0.02	0.00	0.12
C0	WHAM_state_space_unknown_sigmaR	fishing_mortality_se	3000	3000	100	0.01	0.01	0.02	0.00	0.12
C0	WHAM_state_space_known_sigmaR	fishing_mortality_se	3000	3000	100	0.01	0.01	0.02	0.00	0.12

- Runtime comparisons across the 500 simulations revealed marked differences in computational efficiency between software architectures and recruitment configurations (Table 7, Table 8).
  - In the penalized maximum likelihood configuration, TMB-based models were substantially faster than legacy ADMB programs (Table 7):
  - FIMS_penalized_MLE was the fastest model overall, optimizing in a median of 0.19 seconds (mean = 0.24 s, range: 0.10–0.64 s).
  - WHAM_penalized_MLE optimized in a median of 0.60 seconds (mean = 0.58 s).
  - Among ADMB models, ASAP required a median of 0.81 seconds, BAM required 2.16 seconds, and SS3 required 4.76 seconds.
  - In state-space configurations, integrating random-effects recruitment deviations increased optimization time. Rceattle optimized in a median of 1.49–1.56 seconds. FIMS state-space models optimized in a median of 11.83–12.35 seconds (mean = 12.31–12.66 s), outperforming WHAM state-space models, which required a median of 16.20–17.40 seconds (mean = 17.51–19.13 s).

Table 7: Wall-clock parameter optimization run times (seconds) across 500 simulation replicates.

Model Configuration	Replicates	Median (s)	Mean (s)	Min (s)	Max (s)
ASAP	500	0.81	0.82	0.43	1.41
BAM	500	2.16	2.18	1.11	4.10
SS3	500	4.76	4.78	2.42	6.39
FIMS_penalized_MLE	500	0.19	0.24	0.10	0.64
FIMS_state_space_unknown_sigmaR	500	12.35	12.66	3.94	35.55
FIMS_state_space_known_sigmaR	500	11.83	12.31	4.22	27.91
Rceattle_penalized_MLE	495	2.17	2.90	0.83	40.71
Rceattle_state_space_unknown_sigmaR	481	1.56	5.15	1.00	59.93
Rceattle_state_space_known_sigmaR	477	1.49	6.34	1.01	59.94
WHAM_penalized_MLE	500	0.60	0.58	0.00	0.60
WHAM_state_space_unknown_sigmaR	500	17.40	19.13	13.80	39.60
WHAM_state_space_known_sigmaR	500	16.20	17.51	12.60	33.60

- Uncertainty calculation times differed according to how derived quantities are reported (Table 8):
  - FIMS_penalized_MLE computed standard errors via TMB::sdreport in a median of 0.13 seconds, WHAM PML required 0.21 seconds, and Rceattle PML required 0.23 seconds.
  - For state-space configurations, FIMS uncertainty reporting took a median of 3.56–3.70 seconds, compared to 2.33–2.37 seconds for WHAM and 5.43–5.48 seconds for Rceattle.
  - In SS3, full uncertainty estimation and file reporting required a median of 17.39 seconds per replicate.

Table 8: Wall-clock run times (seconds) for Hessian inversion and uncertainty reporting (sdreport / covariance evaluation) across 500 simulation replicates.

Model Configuration	Replicates	Median (s)	Mean (s)	Min (s)	Max (s)
ASAP	500	1.75	1.71	0.66	2.10
BAM	500	1.00	0.98	0.00	1.59
SS3	500	17.39	17.15	9.50	19.12
FIMS_penalized_MLE	500	0.13	0.14	0.03	0.38

Table 8: Wall-clock run times (seconds) for Hessian inversion and uncertainty reporting (sdreport / covariance evaluation) across 500 simulation replicates.

Model Configuration	Replicates	Median (s)	Mean (s)	Min (s)	Max (s)
FIMS_state_space_unknown_sigmaR	500	3.70	3.80	0.84	9.44
FIMS_state_space_known_sigmaR	500	3.56	3.62	0.77	11.76
Rceattle_penalized_MLE	495	0.23	0.29	0.10	0.90
Rceattle_state_space_unknown_sigmaR	483	5.48	6.56	4.12	16.06
Rceattle_state_space_known_sigmaR	477	5.43	6.25	4.04	16.02
WHAM_penalized_MLE	500	0.21	0.22	0.12	0.49
WHAM_state_space_unknown_sigmaR	500	2.37	2.56	1.98	7.00
WHAM_state_space_known_sigmaR	500	2.33	2.46	1.91	6.03

Discussion

- FIMS produced estimates of spawning biomass, recruitment, and fishing mortality that were highly consistent with those from established stock assessment models, indicating comparable estimation accuracy across metrics of interest.
- The similarity in performance across models suggests that, when configured equivalently, differences in underlying software frameworks (e.g., ADMB vs. TMB) have minimal impact on estimation outcomes. This consistency is particularly important for ongoing transitions in assessment platforms, as it indicates that adopting FIMS is unlikely to introduce systematic differences in key management quantities.
- Small differences among models likely resulted from implementation details such as recruitment deviation constraints (e.g., sum-to-zero assumptions) and approaches to equilibrium initialization of numbers-at-age (working in progress).
- Random-effects configurations in both FIMS and WHAM generally reduced bias relative to fixed-effects models. This finding underscores the value of incorporating random effects in stock assessment models. The strong performance of FIMS in this context demonstrates its capability to flexibly implement and estimate random-effects structures.

- These results support the use of FIMS as a reliable and flexible framework for next-generation fisheries stock assessments while maintaining consistency with existing modeling approaches.
- Conducting an objective, "apples-to-apples" benchmarking comparison across six assessment programs required standardizing several structural options and numerical settings so that differences in model behavior reflected fundamental framework formulations rather than divergent auxiliary assumptions:
  - Standardization of Optimizer Loops and Newton Steps in TMB Models. Different TMB-based packages implement varying default convergence protocols in their outer optimization routines. While `FIMS::fit_fims` and `Rceattle::fit_mod` support restart loops (`loopnum` or `number_of_loops`) to re-check gradients after initial convergence, WHAM (`wham::fit_wham`) does not implement an outer loop restart mechanism, relying instead on `nlminb` optimization supplemented by additional Newton steps (`n.newton = 3`) to refine the solution.
  - To ensure an equivalent optimization protocol across all three TMB models, we standardized the optimizer controls across frameworks by setting the number of outer loop restarts to zero in the other two models (`number_of_loops = 0` in FIMS; `loopnum = 1` without restarts in Rceattle) and standardizing the number of Newton steps to three across FIMS, Rceattle, and WHAM. This ensured that all three TMB frameworks were evaluated under identical numerical optimization criteria.
  - Stock-Recruitment Parameterization: Steepness vs. Alpha–Beta. In the operating model and ADMB programs (ASAP, BAM, and SS3), Beverton–Holt recruitment is parameterized using unfished recruitment (R_0) and steepness (h). In contrast, WHAM and Rceattle parameterize the Beverton–Holt relationship using α and β formulation:
$$R = \frac{\alpha SSB}{1 + \beta SSB}$$
. To align WHAM and Rceattle with the operating model's fixed steepness ($h = 0.75$), we analytically calculated α and β from the operating model's true steepness, unfished recruitment (R_0), and unfished spawning biomass per recruit (ϕ_0):
$$\alpha = \frac{4h}{\phi_0(1-h)}, \quad \beta = \frac{\alpha - 1/\phi_0}{R_0}$$
. In WHAM, $\log \alpha$ was fixed to the value implied by the OM steepness (`log_SR_a` mapped to `NA`), estimating only β. Similarly, in Rceattle, we fixed α to the OM-derived mean while estimating β. FIMS directly estimates R_0 while holding steepness constant at $h = 0.75$. This parameter alignment ensured that all six estimation models shared equivalent underlying spawner–recruit dynamics.
- Population Initialization (N_1) and Parameter Dispersion Small differences among models in parameter dispersion across simulations (such as the slight difference in `sd_mre` for

fishing mortality: 3.05%–3.07% in FIMS vs. 2.95%–2.99% in legacy models) resulted from implementation details regarding equilibrium initialization of numbers-at-age:

  - Equilibrium Offset (ASAP, BAM, SS3): ADMB models calculate year-1 abundance by applying an equilibrium age structure based on an initial equilibrium fishing mortality rate (F_1) combined with initial recruitment deviations or offsets. This structural constraint enforces an equilibrium distribution across age classes that dampens cohort variability across early years.
  - Free Parameters (FIMS, WHAM, Rceattle): In contrast, FIMS estimates initial numbers-at-age as unconstrained free parameters (`log_init_naa` across all 12 age classes), matching the unconstrained options in WHAM (`N1_model = 0`) and Rceattle (`initMode = 0`). Estimating 12 independent cohort abundances removes equilibrium assumptions and grants the model full flexibility to fit early age compositions, but naturally introduces slightly wider parameter dispersion across stochastic replicates where early cohort observations contain observation error.

- Random-effects configurations in FIMS, WHAM, and Rceattle generally reduced estimation bias relative to penalized fixed-effects models, underscoring the substantial value of incorporating state-space random effects in stock assessment models:

  - In penalized fixed-effects configurations (ASAP, BAM, SS3, FIMS PML, Rceattle PML, and WHAM PML), spawning biomass exhibited a slight negative median relative error (-0.03% to -0.15%), and cumulative 30-year biomass was underestimated by an average of -0.21%. Transitioning to state-space random-effects formulations reduced median biomass bias to within $\pm$0.01% to +0.04%, and reduced cumulative biomass relative error to -0.05%. Integrating over the recruitment process error distribution via the Laplace approximation accounts for the full probability density of lognormal deviations, naturally mitigating the retransformation bias inherent in penalized fixed-effect deviation vectors.
  - In WHAM and Rceattle, state-space formulations achieved higher convergence rates than their penalized maximum likelihood counterparts (WHAM: 100% vs. 73.0% in PML; Rceattle: 96.2% vs. 87.8% in PML). Under fixed-effects estimation, estimating 29 annual recruitment deviations as independent parameters increases the dimensionality of the gradient vector evaluated by the outer quasi-Newton optimizer (`nlminb`), which can elevate the frequency of saddle-point terminations. Under random effects, these deviations are solved within the inner Laplace approximation, presenting a lower-dimensional, better-conditioned marginal objective function to the outer optimizer.
  - Across FIMS, WHAM, and Rceattle, estimating the recruitment standard deviation (σ_R) freely versus fixing it at the operating model's true value yielded nearly identical point estimates, error distributions, and convergence rates. This confirms that modern TMB assessment models can reliably estimate recruitment process variance without compromising trajectory estimation when composition and survey data are informative.

  - The consistent performance of FIMS across both state-space scenarios demonstrates its capability to flexibly implement, estimate, and integrate random-effects structures.

- While runtime were recorded across all models, direct comparisons of computational run times between ADMB-based executable programs (ASAP, BAM, SS3) and TMB-based R packages (FIMS, WHAM, Rceattle) must be interpreted with caution due to several fundamental architectural differences:

  - ADMB models run as standalone compiled executables (`.exe`). In simulation benchmarks running on Linux workstation or cluster environments, executing ADMB programs requires compatibility translation layers (e.g., Wine) and extensive disk I/O, writing and parsing intermediate ASCII text files (`.dat`, `.par`, `.rep`, `.std`, `admodel.cov`, `Report.sso`) for every replicate. In contrast, TMB packages execute pre-compiled C++ shared objects directly in memory within the R session, eliminating disk latency.
  - The computational workload evaluated during optimization differs considerably across frameworks. SS3 evaluates generalized dynamics, age-length keys, and complex penalty functions that are absent in single-purpose models. For uncertainty reporting, SS3 computes full covariance matrices across dozens of derived biological quantities, whereas TMB models evaluate standard errors only for variables explicitly flagged via `ADREPORT()` or requested in `TMB::sdreport`.
  - In penalized maximum likelihood, TMB models (FIMS PML: 0.19 s; WHAM PML: 0.60 s) were faster than ADMB models because automatic differentiation in CppAD rapidly computes analytic derivatives. However, in state-space configurations, TMB models must repeatedly solve an inner optimization problem to compute the Laplace approximation at every outer iteration. Consequently, state-space run times reflect the computational cost of integrating latent random variables—a functionality not present in classical ADMB statistical catch-at-age models.

- These results support the use of FIMS as a reliable, computationally efficient, and flexible framework for next-generation fisheries stock assessments while maintaining consistency with existing modeling approaches. The modular C++ architecture of FIMS decoupled model configuration from core dynamics, allowing seamless transitions between penalized fixed effects and state-space random effects without altering source code.

- Future benchmark investigations should build upon this baseline evaluation to examine model behavior under more challenging assessment conditions. The multi-model comparison framework developed here provides a reproducible testbed for evaluating these emerging assessment challenges.

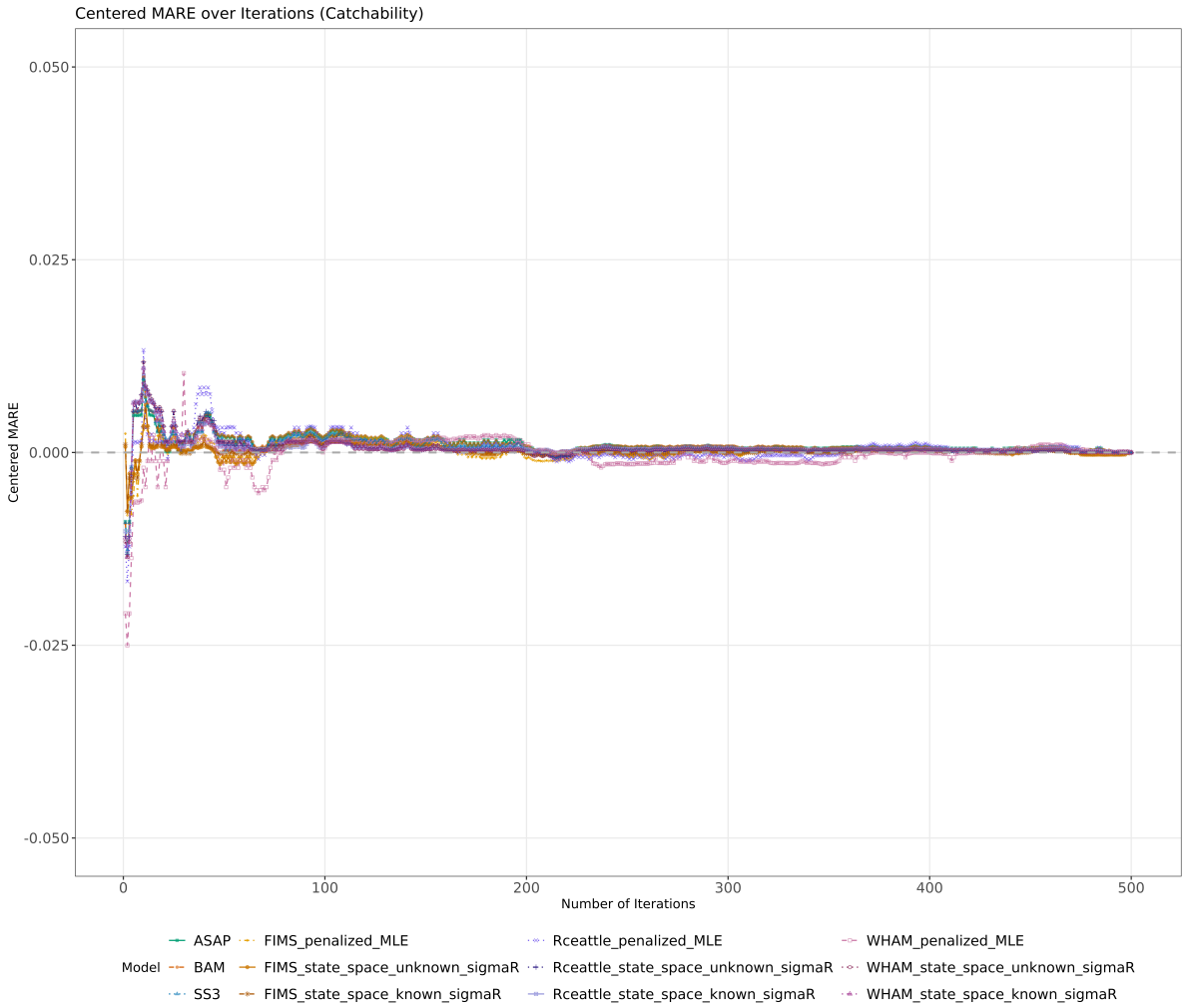

Figure 4: Centered Median Absolute Relative Errors (MAREs) ($MARE_i - MARE_{500}$, where i = number of simulations used in calculation) of catchability (q) versus the number of replicates included in the calculation. Results are shown for seven estimation models: ASAP, BAM, SS3, WHAM (fixed and random effects), and FIMS (fixed and random effects).

Appendix

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